

Face Is Not All You Need: MIME Benchmark for Incomplete Multimodal Emotion Recognition

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Abstract

Multimodal Emotion Recognition (MER) is a fundamental task in multimedia understanding, where state-of-the-art methods have achieved prominent success relying on the assumption of complete, well-aligned multimodal inputs. However, real-world unconstrained scenarios often suffer from unpredictable modality degradation and information loss, leading to catastrophic performance collapse of existing MER models that heavily depend on facial cues as the core inference anchor. Worse still, there remains a critical research gap: few systematic works exist to evaluate model robustness under diverse, fine-grained incomplete information settings, as prior works are limited to coarse-grained full-modality removal and neglect interpretable analysis of emotional reasoning processes. To bridge this gap, we formally define the Incomplete Multimodal Emotion Recognition (IMER) task, which benchmarks model generalization and robustness under multi-grained modality and emotional information loss. We further construct MIME, a dedicated IMER benchmark with 2,000 video segments covering realistic fine-grained facial-detail degradation, severe facial degradation, and extreme full-modality loss, surpassing the limited coarse-grained settings of prior works. We also propose the Chain of Emotion (CoE) analysis paradigm with tailored evaluation metrics to dissect the fine-grained emotional cue perception and reasoning process of Multimodal Large Language Models (MLLMs). Extensive experiments on seven mainstream MLLMs reveal that body language and scene context act as critical compensatory mechanisms under facial degradation, while a pronounced gap exists between models' basic visual perception ability and high-level cognitive reasoning. Our work provides a focused evaluation platform and actionable insights for advancing robust multimodal emotion recognition. Our dataset and code are public at <https://yuxinokk.github.io/MIME>.

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CCS Concepts

• Security and privacy → Social network security and privacy.

Keywords

Multimodal Emotion Recognition, Missing Modalities, Chain-of-Emotion, Benchmark Dataset, Large Vision-Language Models

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1 Introduction

Multimodal Emotion Recognition (MER) is an important task in multimedia, aiming to infer human emotional states from multimodal information [9, 14, 17, 42]. Recent advancements in MER [1, 20, 24] have demonstrated remarkable success by effectively fusing diverse multimodal information, such as facial dynamics, vocal nuances, body language, and contextual visual cues, etc, which usually rely on the assumption of completed and well-aligned multimodal inputs.

However, in unconstrained real-world deployment scenarios, the acquired multimodal data often suffers from unpredictable degradation and modality missing due to various practical factors [7, 25, 39, 49], such as severe facial occlusion, lighting fluctuations, sensor malfunctions, network packet loss, and privacy constraints. Most existing MER methods [19, 23, 40, 43] heavily rely on facial expression features as the core and dominant cue for emotion inference, which leads to catastrophic performance degradation or even complete functional failure when the critical facial modality is corrupted, invalidated, or entirely unavailable.

Despite the urgent demand for robust MER systems [5, 13, 36] in practical scenarios, there remains a notable and unexplored research gap: few works have systematically investigated and benchmarked how current MER models perform under diverse, fine-grained, incomplete information settings. To bridge this gap, we formally define a novel task: **Incomplete Multimodal Emotion Recognition (IMER)**, which focuses on evaluating the generalization ability

and robustness of emotion recognition models under varying levels of modality and emotional information missing. Compared with the standard MER task built on ideal complete-modal assumptions, IMER can more accurately reflect the real-world performance and practical reliability of emotion recognition systems in unconstrained deployment environments.

To facilitate the proposed task, we construct **Missing Information in Multimodal Emotion (MIME)**, a dedicated benchmark tailored for assessing model performance under diverse information-missing scenarios, which contains 2,000 video segments. Unlike existing IMER benchmarks that only simulate extreme, coarse-grained full-modality missing (e.g., complete absence of audio or visual streams), MIME emphasizes that real-world incompleteness is inherently multi-grained: in many practical cases, the entire visual modality is not fully lost, but the critical facial emotional evidence is only partially degraded. We therefore distinguish fine-grained facial-detail degradation from more severe facial degradation, rather than assuming that facial evidence is either fully available or entirely absent.

In addition, we design a novel analysis paradigm termed Chain of Emotion (CoE), to dissect the fine-grained emotional cue perception and reasoning process of Multimodal Large Language Models (MLLMs) during emotion recognition. Specifically, CoE explicitly guides models to decouple and articulate their observations across individual modalities—namely face, body, scene, and audio—prior to predicting the final emotion, moving beyond the single final accuracy metric to provide interpretable insights into model behaviors. Furthermore, we develop task-specific evaluation metrics for CoE, which are designed for scene understanding, emotional analysis, classification Accuracy and overall CoE quality.

Our extensive evaluation results reveal that while the face is the primary emotional indicator, body language and scene context act as essential compensatory mechanisms under facial degradation (effectively mitigating catastrophic performance collapse). Even the most powerful MLLMs suffer from significant performance drops when facing fine-grained facial information corruption and cross-modality joint missing scenarios, exposing severe robustness limitations of existing models in real-world settings. CoE evaluations expose a pronounced gap between the models' fundamental visual perception capabilities and their high-level cognitive reasoning skills. Our dataset and code are available at <https://github.com/oiiizhou/MIME>.

The main contributions of our work are summarized as follows:

- We formally define the IMER task, the first systematic task formulation focusing on multi-grained information missing in MER, which bridges the long-standing gap between standard ideal settings and real-world deployment demands.
- We construct MIME, the first dedicated benchmark for IMER, equipped with the CoE analysis paradigm and task-specific evaluation metrics, providing a standardized and extensible evaluation platform for the research community.
- We conduct comprehensive benchmark experiments on 7 mainstream open-source and closed-source MLLMs under various missing-modality conditions. We reveal their strengths and limitations, paving the way for future research efforts to advance emotional intelligence in MLLMs

Table 1: Comparison of MIME with other Benchmarks related to IMER.

Benchmark	Format	Supp. Incomplete	Incom. grained	Chain of Emotion
EmotionBench [11]	Text	✗	✗	✗
EmoBench [35]	Text	✗	✗	✗
MOSABench [37]	Image	✗	✗	✗
MM-InstructEval [45]	Image	✗	✗	✗
IEMOCAP [2]	Video	✓	Coarse	✗
MC-EIU [26]	Video	✗	✗	✗
OV-MER [21]	Video	✗	✗	✗
EmoBench-M [10]	Video	✗	✗	✗
MER-UniBench [22]	Video	✗	✗	✗
MOSI [47]	Video	✓	Coarse	✗
MME-Emotion [48]	Video	✗	✗	✓
MIME (Ours)	Video	✓	Fine & Coarse	✓

2 Related Work

2.1 Multimodal Emotion Recognition Datasets

The evolution of Multimodal Emotion Recognition (MER) has been fundamentally driven by large-scale datasets. Early affective computing research relied on laboratory-collected datasets like IEMOCAP [3] and MELD [32]. To capture more naturalistic human emotions, recent benchmarks have shifted toward *in-the-wild* scenarios. For instance, CAER [16] pioneered context-aware emotion recognition by emphasizing the joint contribution of facial expressions and background scenes. DFEW [12] and MAFW [27] provided large-scale dynamic video clips extracted from real-world media, encompassing rich audio-visual multimodal cues. Furthermore, FERV39k [41] contributed a massive collection of unconstrained, silent video clips, driving the research of visual-only emotion analysis. However, a fundamental limitation of these benchmarks is their reliance on the assumption of modality completeness. Furthermore, their annotations are strictly limited to discrete categorical labels (e.g., *Happy*, *Sad*). They lack the fine-grained, language-based reasoning rationales required to interpret complex emotional states [39, 44, 46, 49], significantly limiting their utility in training interpretable affective models.

2.2 Emotion Recognition under Missing Modalities

Recent works [8, 18, 33] explore MER under missing modalities, but they are limited to a coarse-grained setting that simply removes fixed modalities from standard datasets and retains only 1-2 modalities. This paradigm cannot support evaluation under semantically specific information loss, nor can it model emotional reasoning processes or analyze the synergistic effect of effective modalities on emotion recognition [50].

To address this gap, we build MIME benchmark with multiple granularities for missing modalities. Unlike previous datasets, our benchmark uniquely decouples visual elements into distinct fine-grained modalities and supports comprehensive missing-modality evaluation for robustness.

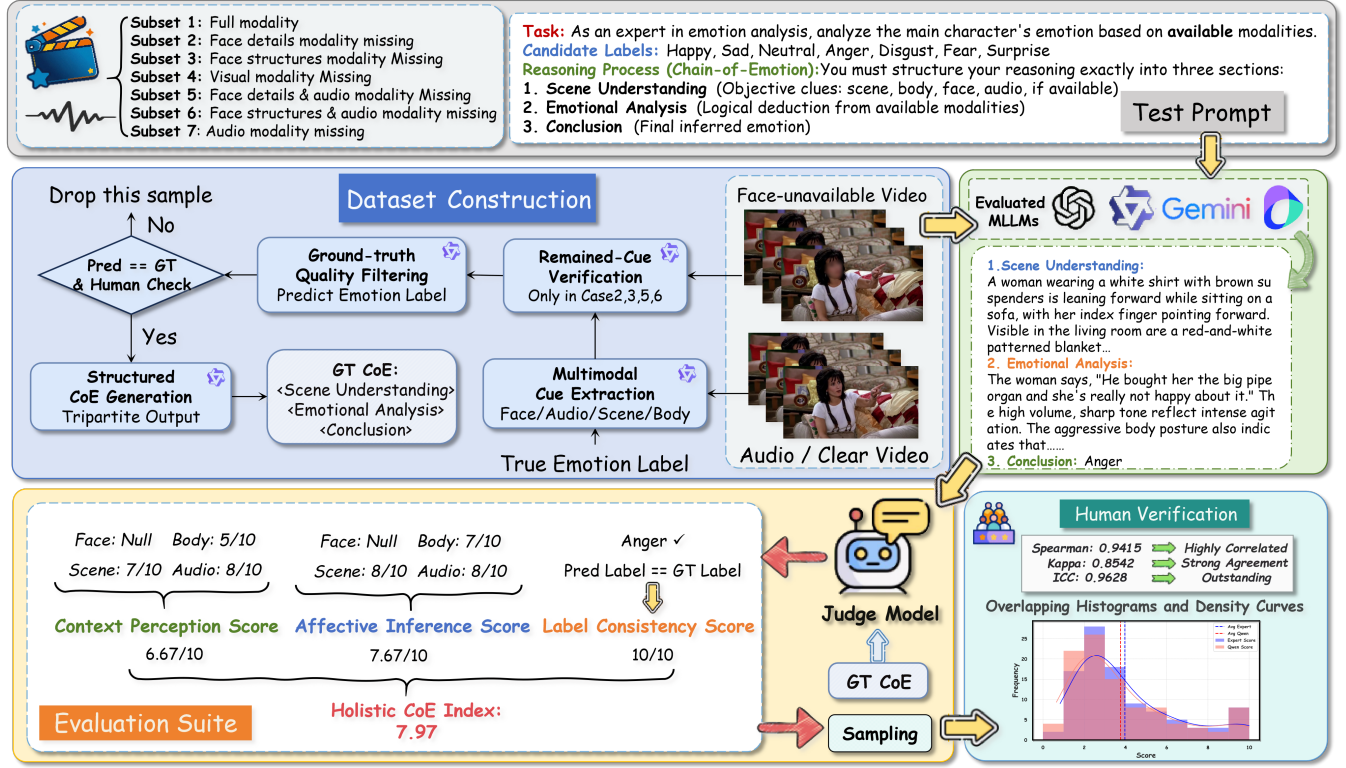


Figure 1: Illustration of Our Data Construction and Evaluation Strategy.

3 MIME Benchmark

3.1 Data Acquisition and Preprocessing

To ensure diversity in social contexts and missing levels, we collect raw data from social media and four authoritative *in-the-wild* benchmarks: CAER, DFEW, MAFW, and FERV39k. We focus on seven universal emotion categories: *Happy*, *Sad*, *Neutral*, *Anger*, *Disgust*, *Fear*, and *Surprise*. A balanced resampling strategy is applied to mitigate long-tail category bias, ensuring a rigorous evaluation of models' generalizability.

Through the modal degradation technique, we systematically decouple video signals into seven subsets to simulate real-world incompleteness: Full modality (FM), Face details missing (FDM), Face structures missing (FSM), Visual modality missing (VMM), Face details and audio modality missing (FDAM), Face structures and audio modality missing (FSAM), Audio modality missing (AMM).

3.2 Chain-of-Emotion (CoE) Generation

To endow our dataset with rich, fine-grained reasoning annotations, we design a comprehensive four-stage pipeline utilizing **Qwen3-Omni** to generate high-quality Chain-of-Emotion. As shown in Fig.1, this pipeline explicitly guides models to decouple and articulate their observations across individual modalities prior to predicting the final emotion. The detailed information of the generation process can be found in the appendix. The logic flow of CoE is divided into three folds:

- **Scene Understanding (C_{scene}):** Characterizing the objective multimodal cues present in the input, extracting factual information exclusively from the available modalities (face, body, scene, and audio).
- **Emotional Analysis (A_{emo}):** Executing cognitive reasoning to map the previously identified cues to internal emotional states. This step explicitly articulates the logical path of how the observed signals lead to the emotional deduction.
- **Predicted Emotion (L):** Deriving the final discrete emotion category, which strictly aligns with the ground-truth label, based on the preceding systematic analysis.

3.3 Statistical Analysis

To ensure the robustness and generalizability of the proposed benchmark, we conduct a comprehensive statistical analysis of the constructed MIME dataset, which contains 2,000 segments. Fig.2 and Fig.3 presents an **Overview of MIME Statistics**. As illustrated in the **Task Types (Fig.2)**, the dataset exhibits a hierarchical distribution: it encompasses specific modality missingness types (inner circle, e.g., Audio Miss, Heavy degeneration Face, Visual Miss) systematically decoupled across diverse *in-the-wild* scene contexts (outer circle, e.g., Lifestyle, Movie, Vlog), enabling fine-grained analysis of diverse video contexts. Fig.3-Top demonstrates that the dataset rigorously adheres to a balanced sample distribution across the seven universal emotion categories, effectively mitigating the long-tail bias prevalent in *in-the-wild* datasets. Furthermore, as

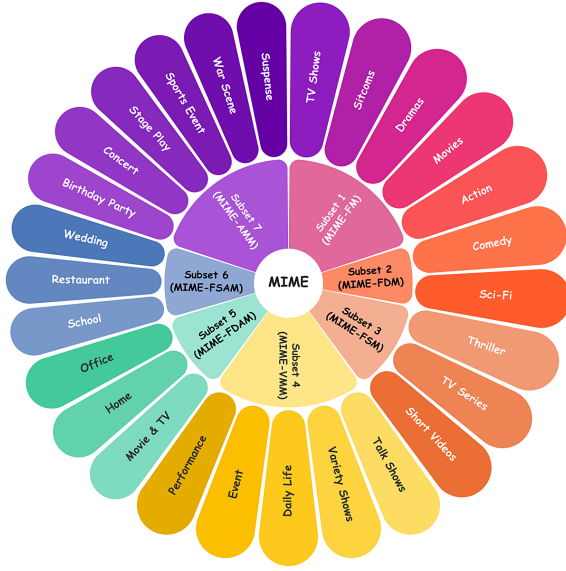


Figure 2: Overview of MIME's Task Types. Task types across diverse video scenarios.

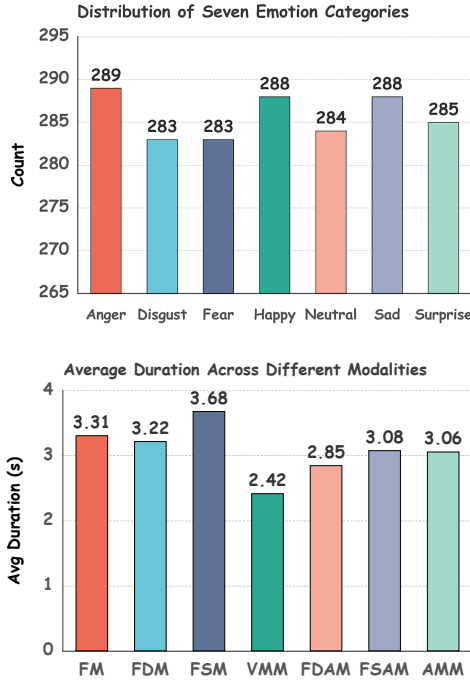


Figure 3: Balanced distributions of emotion categories(top) and modality-specific video durations(bottom).

shown in the Fig.3-Bottom, the average durations of the audio/video data across different modality combinations are evenly distributed, facilitating a comprehensive evaluation of temporal understanding under various degraded cases.

3.4 Evaluation Suite

Evaluation Strategy. Traditional evaluation frameworks predominantly rely on discrete emotion labels, which struggle to comprehensively measure the reasoning quality of models in complex, modality-impaired scenarios. Inspired by the annotation-free evaluation paradigms introduced in recent emotional intelligence benchmarks (e.g., MME-Emotion), we introduce a hybrid evaluation strategy grounded in the LLM-as-a-Judge mechanism. Specifically, we deploy Qwen3.5-plus as a judge model to conduct automated and fine-grained quantitative scoring on the CoE generated by the evaluated models. This approach not only evaluates the final classification accuracy but also objectively measures the logical rigor and factual basis of the models' subjective reasoning processes, all while circumventing prohibitive manual annotation costs.

Evaluation Metrics. To comprehensively dissect the emotional intelligence of MLLMs from both objective classification performance and subjective interpretability dimensions, we compartmentalize our evaluation metrics into the following two core modules:

Stage I: Standard Classification metrics

For the final emotion prediction, we compute the **Accuracy (Acc)** alongside **F1-score** to rigorously evaluate the models' classification robustness across the seven emotion categories, particularly mitigating potential long-tail distribution issues under modality degradation.

Stage II: CoE Quality Metrics

To meticulously quantify the "compensatory reasoning" capability, the LLM-as-a-Judge conducts fine-grained scoring on the generated CoE based on four predefined modalities: Face (f), Body (b), Scene (s), and Audio (a). We denote the modality set as $\mathcal{M} = \{f, b, s, a\}$, and let $I_m \in \{0, 1\}$ indicate whether modality m is valid for scoring, where $I_m = 1$ if modality m is available or degraded-but-observable, and $I_m = 0$ if modality m is truly missing. The evaluation yields three sub-metrics and one holistic index:

Context Perception Score (CPS): Evaluates whether the model accurately captures the surviving physical cues. The overall CPS is computed over the valid modalities only, where $s_m^{cps} \in [0, 10]$:

$$CPS = \frac{\sum_{m \in \mathcal{M}} I_m \cdot s_m^{cps}}{\sum_{m \in \mathcal{M}} I_m} \quad (1)$$

Affective Inference Score (AIS): Assesses the logical rigor of deducing emotional states from the perceived cues of each modality, explicitly penalizing hallucination, where each valid modality receives a score $s_m^{ais} \in [0, 10]$:

$$AIS = \frac{\sum_{m \in \mathcal{M}} I_m \cdot s_m^{ais}}{\sum_{m \in \mathcal{M}} I_m} \quad (2)$$

Label Consistency Score (LCS): Checks whether the semantic conclusion of the long-text reasoning path strictly aligns with the final categorical prediction. LCS equals 10 if correct, otherwise 0.

Holistic CoE Index (HCI): Serves as the ultimate benchmark for measuring the model's comprehensive "affective interpretability." It is the weighted integration of the above three metrics:

$$HCI = \alpha \cdot CPS + \beta \cdot AIS + (1 - \alpha - \beta) \cdot LCS \quad (3)$$

where the hyperparameters are empirically set as $\alpha = 0.4$ and $\beta = 0.3$ to emphasize foundational context understanding.

Table 2: Overall Performance Comparison (%) on MIME. The top three performing results are highlighted in red (1st), blue (2nd), and yellow (3rd).

Model	F1-Score	CPS	AIS	LCS	HCI
Qwen3.5-27b [34]	40.99	6.57	5.13	4.32	5.46
Qwen3.5-plus [34]	42.60	6.70	5.32	4.43	5.60
GPT-5.4-nano [31]	26.21	5.41	3.97	2.93	4.23
Gemini-3.1-flash [6]	46.61	6.54	5.24	4.68	5.59
GPT-5-mini [29]	33.52	5.91	4.42	3.48	4.73
Doubao-seed-2-0 [4]	45.27	6.43	4.96	4.65	5.46
GPT-5-nano [30]	27.01	5.53	4.06	2.95	4.31

Human Verification. To validate the reliability of our proposed MLLM-as-judge framework, we conducted manual evaluation with 3 expert annotators. The experts annotated 173 samples and corresponding CoE outputs to analyze the alignment between Qwen-generated scores and human annotations. As shown in the bottom-right panel of Fig. 1, the two evaluations achieve a Spearman’s rank correlation coefficient [38] of 0.9415 (strong positive correlation), a Cohen’s Kappa coefficient [28] of 0.8542 (near-perfect inter-rater agreement), and an intra-class correlation coefficient (ICC) [15] of 0.9628 (outstanding grading consistency). These metrics confirm that our MLLM-as-judge methodology closely mirrors human assessment, verifying its credibility and robustness as an evaluation approach.

4 Experiments

4.1 Experimental Setup

We evaluate the performance on a total of 7 cutting-edge MLLMs, including the GPT-5 series (gpt-5.4-nano, gpt-5-mini, gpt-5-nano), the Gemini series (gemini-3.1-flash), the Qwen suite (qwen3.5-plus, qwen3.5-27b), and doubao-seed-2-0. All the experiments are strictly conducted under zero-shot settings. Furthermore, to achieve scalable qualitative evaluation, we employ Qwen3.5-Plus as the primary LLM-as-a-Judge. Leveraging a concurrent execution architecture with multi-threading, the judge model rigorously scores the generated CoE texts based on their logical alignment and factual grounding.

4.2 Main Result

We comprehensively showcase the performance comparison of various models on the MIME benchmark in Tab.2 and 3, encompassing both standard classification metrics (e.g., ACC and WF1) and CoE Quality Metrics scored by the judge agent.

Overall Classification and Reasoning Performance: The experimental landscape clearly reveals the immense challenge of multimodal emotion recognition under modality-impaired scenarios. Even the top-performing model, gemini-3.1-flash, achieves F1-Score of only 46.61 and an HCI of 5.59. It is closely followed by doubao-seed-2-0 (F1: 45.27, HCI: 5.46) and qwen3.5-plus (F1: 42.60, HCI: 5.60). In stark contrast, smaller-scale models such as gpt-5.4-nano and gpt-5-nano experience a precipitous performance drop, with F1-Score plummeting to 26.21 and 27.01, respectively, and

their HCI hovering around 4.23 and 4.31. The fact that all evaluated models report F1-Score below 50 indicates that even state-of-the-art MLLMs are still in the nascent stages of developing robust emotional intelligence for complex, in-the-wild environments.

The Gap Between Context Perception and Affective Inference: Delving into the CoE quality metrics, we observe a pronounced divergence between "Scene Understanding (CPS)" and "Emotional Analysis (AIS)." For instance, while qwen3.5-plus achieves a high score of 6.70 in perceiving foundational physical cues (CPS), its score drastically drops to 5.32 when translating these cues into internal affective inference (AIS). Similarly, gpt-5-mini scores 5.91 and 4.42 on these two metrics, respectively. This phenomenon profoundly exposes the affective cognitive bottleneck of current MLLMs: they are capable of adequately "seeing" and extracting objective physical elements in degraded environments, but they severely lack the capacity to execute deep, compensatory logical reasoning based on those observations.

Vulnerability Under Diverse Missing Conditions: Further observing the performance across the seven specific modality conditions (Subsets 1–7), we note significant classification instability. Most models exhibit drastic performance fluctuations when confronted with different combinations of missing modalities. Taking gpt-5-mini as an example, its accuracy reaches 50.00% under Subset 4 (Visual Modality Missing), yet plummets to a mere 24.68% in Subset 2 (Face Details Modality Missing) (a steep decline of 26%). Such severe dependency on specific explicit modalities (e.g., intact and clear facial features) forces models to degenerate into distribution-based "probabilistic guessing" when key visual cues are distorted or absent, as they fail to effectively utilize remaining modalities, body language, or scene context as compensatory evidence. This drastically constrains their zero-shot generalization capabilities in open-world settings.

4.3 Observations and Insights

Through cross-analysis of the quantitative classification results and qualitative CoE evaluations, we distill three insights regarding MLLMs’ affective reasoning under modality-impaired conditions (Subsets 1–7):

Obs. 1: A Pronounced Gap Between Visual Perception and Cognitive Reasoning. "Seeing" a scene does not guarantee "understanding" its emotion. Across all models, CPS (Context Perception Score) consistently exceeds AIS (Affective Inference Score). For instance, qwen3.5-plus accurately extracts physical cues (CPS: 6.70/10), but struggles to logically deduce internal emotional states (AIS: 5.32/10). This indicates that while MLLMs possess strong visual foundations, they lack human-like empathic cognition to bridge the semantic gap from objective physical space to subjective affective space.

Obs. 2: Missing Key Modalities Triggers "Shortcut Dependency" and Category Collapse. When explicit features degrade (e.g., missing facial details or structures in Subset FDM, FSM, FDAM, FSAM), models fail to utilize compensatory cues like body language, relying instead on pre-training data biases. This causes catastrophic performance collapse in categories heavily dependent on facial micro-expressions, like "Disgust". Specifically, gpt-5.4-nano yields a dismal F1-score of 0.16 for Disgust, and doubao-seed-2-0 achieves

Table 3: Classification performance under different subsets.

Model	FM		FDM		FSM		VMM		FDAM		FSAM		AMM	
	ACC	WF1	ACC	WF1	ACC	WF1	ACC	WF1	ACC	WF1	ACC	WF1	ACC	WF1
gpt-5.4-nano [31]	22.81	18.92	25.95	19.70	31.12	27.80	43.50	42.28	20.67	16.29	31.94	29.72	25.00	21.03
gpt-5-mini [29]	27.07	24.13	24.68	22.80	36.93	34.70	50.00	48.78	23.08	21.98	44.27	42.36	31.75	31.64
gemini-3.1-flash [6]	46.10	44.83	50.96	49.82	49.38	50.09	44.25	42.75	37.98	37.31	48.42	47.33	50.75	51.34
gpt-5-nano [30]	23.75	19.87	20.89	16.48	28.22	25.22	49.50	47.32	19.23	16.61	35.94	33.47	21.75	17.98
qwen3.5-plus [34]	48.12	45.32	44.94	42.26	40.25	38.79	42.50	39.87	35.58	32.30	50.26	49.28	46.00	44.44
qwen3.5-27b [34]	44.11	40.57	48.73	46.81	42.74	40.79	42.61	39.48	35.10	30.75	46.35	44.43	43.50	40.64
doubao-seed-2-0 [4]	51.13	48.10	48.73	48.29	48.13	45.61	41.50	39.24	40.87	38.02	47.92	47.65	47.12	45.72

Table 4: Per-emotion Accuracy (ACC) and F1-score (F1) comparison of different models.

Model	Happy		Sad		Neutral		Anger		Disgust		Fear		Surprise	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
gpt-5.4-nano [31]	15.28	0.23	26.04	0.27	33.10	0.28	53.50	0.41	9.19	0.16	29.72	0.35	8.42	0.14
gpt-5-mini [29]	21.18	0.28	33.68	0.36	51.41	0.31	57.14	0.43	11.66	0.20	45.23	0.45	23.51	0.30
gemini-3.1-flash [6]	50.69	0.57	53.31	0.52	68.20	0.41	56.14	0.52	21.20	0.32	37.81	0.46	40.14	0.47
gpt-5-nano [30]	12.15	0.19	24.31	0.29	68.66	0.31	35.07	0.39	4.59	0.09	45.58	0.38	16.49	0.24
qwen3.5-plus [34]	63.76	0.59	56.25	0.48	50.00	0.35	56.45	0.54	7.77	0.14	42.4	0.48	32.63	0.40
qwen3.5-27b [34]	66.67	0.58	55.56	0.45	43.66	0.35	50.87	0.49	6.71	0.12	42.05	0.45	35.92	0.43
doubao-seed-2-0 [4]	68.75	0.59	48.61	0.50	57.24	0.40	57.84	0.53	17.31	0.27	31.45	0.41	43.51	0.46

only 0.27. Such extremely low recall confirms that occluded visual/facial modalities severely impair specific emotion recognition, exposing the fragility of models in open-world scenarios.

Obs. 3: Explicit CoE Quality Dictates Predictions in Complex Scenarios. Our analysis reveals a strong positive correlation between objective classification accuracy and the overall quality of Compensatory CoE reasoning, as measured by the HCI metric. State-of-the-art models (e.g., gemini-3.1-flash, qwen3.5-plus, doubao-seed-2-0) consistently maintain HCI scores above 5.45. In contrast, lightweight models such as the GPT-nano series, which yield F1-scores below 30, exhibit fragmented or superficial reasoning traces and score under 4.40 on HCI. Crucially, the leading models also achieve higher Label Consistency Scores (LCS), indicating that their final predictions are grounded in stable, step-wise logical deduction rather than heuristic shortcuts or random guessing. This consistency demonstrates that high-quality CoE reasoning acts as a reliability anchor when multimodal cues are incomplete or degraded. Consequently, systematically reinforcing MLLMs’ internal “Compensatory Chain-of-Emotion” emerges as a highly promising paradigm for mitigating multimodal affective bottlenecks.

5 Conclusion

In this paper, we address a critical underexplored challenge in affective computing: multimodal emotion recognition under missing modalities. We further propose MIME, the first fine-grained Chain-of-Emotion benchmark. It systematically simulates real-world modality degradation across seven scenarios to build a rigorous testing environment. Built via an adaptive four-stage generation pipeline



Task: As an expert in emotion analysis, analyze character's emotion based on available modalities.
Candidate Labels: Happy, Sad, Neutral, Anger, Disgust, Fear, Surprise
Ground Truth CoE:

- Scene Understanding:** The young individual with long dark hair and wearing a patterned headscarf, a red vest over a striped shirt, and a white outer garment stands tense in a dim, grassy field at twilight. The face is heavily blurred. Her head tilts upward, arms close, suggesting alertness amid eerie lighting.
- Emotional Analysis:** Although the face is heavily blurred, we can still infer fear from the following cues. The young individual says "What... what was that?" with a high-pitched, trembling, rapid tone. The dim, eerie scene amplifies unease. Her tense posture and upward head tilt reinforce apprehension.
- Conclusion:** Fear

Qwen3.5-plus:

- Scene Understanding:** The subject is a young girl in a dimly lit, grassy outdoor setting, surrounded by other standing figures. The lighting is cool and blue-tinted, creating a somber atmosphere. The girl stands relatively still with a slight forward lean, while the surrounding figures remain static, establishing a tense, possibly confrontational or ritualistic group context.
- Emotional Analysis:** Despite the facial blur, the girl's body language suggests a state of distress or apprehension. Her slight forward lean and stillness imply hesitation or being overwhelmed. The gloomy, low-key lighting and the encircling presence of others create an oppressive environment. These contextual clues strongly point towards a negative, high-arousal emotional state like fear or anxiety rather than anger or sadness.
- Conclusion:** Fear

GPT-5-mini:

- Scene Understanding:** A child stands outdoors on grass among peers, wearing a headscarf, striped shirt and red vest. The subject is upright and repeatedly tilts the head sharply upward across frames while surrounding figures remain mostly stationary. A sudden bright flare appears at screen right. Movement suggests a reactive, attention-grabbing external event in a group setting.
- Emotional Analysis:** The quick, repeated upward head tilt signals an immediate startle or astonishment response. Lack of retreat, aggressive stance, or defensive gestures among the group reduces likelihood of sustained fear or anger. The transient nature of the motion and the bright flare point to Surprise as the primary inferred emotion.
- Conclusion:** Surprise

Rating

Qwen3.5-plus: 4.67/10

GPT-5-mini: 5/10

GPT-5-nano: 10/10

GPT-5-mini: 4.67/10

GPT-5-nano: 3/10

GPT-5-nano: 0/10

Figure 4: Error cases of model responses.

with conditional hallucination suppression, MIME delivers high-quality tripartite structured reasoning chains. We also design a dual-stage hybrid evaluation suite with an LLM-as-a-Judge, to quantify models’ objective classification robustness and subjective reasoning logic.

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